

Spectral continuity, amplitude changes, and perception of length contrasts

Abstract

Japanese deploys a singleton-geminate contrast in obstruents and nasals, but not in glides. Even though Japanese allows lexical nasal geminates, patterns of emphatic gemination show that Japanese avoids creating nasal geminates. Japanese therefore disfavors sonorant geminates in general, and glide geminates in particular. This paper tests hypotheses about why Japanese shows these preferences. Concerning the distinction between obstruent geminates and sonorant geminates, Podesva (2002) hypothesizes that the phonological dispreference against sonorant geminates exists because these geminates are easily confused with corresponding singletons. This confusability problem arises because sonorants are spectrally continuous with flanking vowels, and consequently their constriction durations are difficult to perceive. Two non-speech perception experiments, Experiments I and II, confirm this hypothesis by showing that length distinctions of consonant intervals that are spectrally continuous with surrounding segments are difficult to perceive. Concerning the difference between nasal geminates and glide geminates, this paper builds on the finding by Kato et al. (1997) that given streams of sounds, listeners use amplitude changes to demarcate segmental boundaries. Experiments III and IV show that amplitude changes facilitate categorization and discrimination of short/long contrasts of consonantal length. This conclusion is compatible with the fact that Japanese disfavor glide geminates more than nasal geminates. Overall the results of the four perception experiments reported here accord well with the phonological patterns of Japanese geminates.

1 Introduction

Japanese uses a lexical singleton-geminate (short vs. long) contrast, and this phonological contrast is primarily cued by a difference in consonantal duration (Kawahara, 2015). This contrast is limited to (voiceless) obstruents and nasals, as shown by the examples in (1) (Kawagoe, 2015). On the other hand, Japanese does not allow lexical singleton-geminate contrasts in glides.¹

(1) Lexical singleton/geminate contrasts in Japanese

¹Japanese lacks geminate of [rr] as well, and we will return to the discussion of liquid geminates in section 2.2.

kata	‘bought’	katta	‘bought’
iso	‘shore’	isso	‘rather’
kona	‘powder’	konna	‘such’

Even though Japanese does allow nasal geminates, one phonological process shows that Japanese avoids creating nasal geminates as well. Nasu (1999) points out that given reduplicated sound-symbolic, mimetic $C_1VC_2V-C_3VC_4V$ forms, in order to create their emphatic forms, Japanese speakers predominantly geminate the second consonant when the consonant is a stop, as in (2).

(2) Emphatic forms via gemination of C_2 when C_2 is a stop

- a. /pata-pata- μ / $\rightarrow$ [pattapata] ‘running’
- b. /pika-pika- μ / $\rightarrow$ [pikkapika] ‘shining’

However, when C_2 is a nasal and C_3 is a stop, speakers prefer to target C_3 , as in (3). In the experiment reported in Kawahara (2013), C_3 was fixed as stops, and C_2 was systematically varied among stops, fricatives and nasals. When asked to create emphatic forms of nonce mimetic words, when C_2 and C_3 are both stops, Japanese speakers chose C_2 -gemination about 80% of the time, supporting the preference in (2). However, when C_2 is a nasal and C_3 is a stop, they chose C_2 -gemination only about 35% of the time and instead resort to C_3 -gemination, as schematized in (3). The flopping of gemination locus in (3) shows that Japanese avoids nasal geminates when possible.

(3) When C_2 is a nasal and C_3 is a stop, speakers prefer C_3 -gemination

- a. /kano-kano- μ / $\rightarrow$ [kanokkano] (nonce word)
- b. /kina-kina- μ / $\rightarrow$ [kinakkina] (nonce word)

Furthermore, in loanword gemination pattern in which word-final consonants in the source languages are borrowed as geminates, oral stops undergo gemination, but nasal stops do not (Katayama, 1998). This asymmetry is shown in (4) and (5):

(4) Gemination of oral stops in loanword adaptation

- a. “stop” $\rightarrow$ [sutoppu]
- b. “top” $\rightarrow$ [toppu]

(5) Nasals do not geminate in the same environment

- a. “Tom” $\rightarrow$ [tomu]
- b. “ham” $\rightarrow$ [hamu]

We acknowledge that loanword adaptation patterns should be used in phonological argumentation with caution (de Lacy, 2006, 2009), because loanword adaptation is non-trivially affected by non-phonological—e.g. perceptual, orthographic, and sociolinguistic—factors (e.g. Kang 2011;

Peperkamp 2005; Peperkamp & Dupoux 2003; Silverman 1992; Takagi & Mann 1994; Vendelin & Peperkamp 2006). At any rate, the asymmetry between (4) and (5) is at least compatible with the view that Japanese avoids nasal geminates, although this argument is admittedly not a very one.

In summary, Japanese avoids glide geminates the most in the sense that it does not allow them to make lexical contrasts. Japanese does allow nasal geminates to signal lexical contrasts, but nevertheless avoids creating them in some gemination process(es). The preferential hierarchy in the phonology of Japanese is therefore: obstruent geminates > nasal geminates > glide geminates. This paper addresses why this hierarchy holds.²

2 The phonetic grounding of the dispreference against sonorant geminates

2.1 The hypothesis

This paper begins by addressing the distinction between obstruent geminates and sonorant geminates. Podesva (2002) proposes that sonorant geminates are dispreferred because they are perceptually confusable with corresponding singletons. The logic goes as follows: sonorants have blurry transitions into and out of flanking vowels, because sonorants are spectrally continuous with surrounding vowels, where “spectrally continuous” means “continuation of periodic energy into and out of surrounding intervals”. It is thus hard to pin down where sonorants begin and where they end

²The same hierarchy holds in other languages, most clearly in Ilokano (Hayes 1989: 270-271). Ilokano resolves hiatus by gliding a first vowel, and this formation of a glide causes compensatory gemination of the preceding consonant. This gemination process usually applies to obstruents as in (6). In the same environment, gemination is marginally possible for nasals as in (7)—according to Hayes, gemination of these consonants is optional, possibly with lexical variation. Gemination never applies to [w, y], as in (8).

- (6) Obstruents usually geminate after gliding of vowels
 - a. /lúto-én/ → [luttwén] ‘cook GOAL-FOCUS’
 - b. /pag-ʔáso-án/ → [pagʔasswán] ‘place where dogs are raised’
 - c. /kina-ʔapó-án/ → [kinaʔappwán] ‘leadership qualities’
 - d. /bági-én/ → [baggyén] ‘to have as one’s own’
 - e. /pag-ʔatáke-án/ → [pagʔatákkýán] ‘place where an attack takes place’
- (7) Nasals only sporadically geminate
 - a. /dámo-én/ → [damwén], ʔ[dammwén] ‘to be new to something’
 - b. /na-ʔalino-án/ → [naʔalinwán], ʔ[naʔalinnwán] ‘to become sensitive’
 - c. /pag-ʔaliŋó-án/ → [pagʔaliŋwán], ʔ[pagʔaliŋŋwán] ‘place where boars are found’
- (8) Glides never geminate
 - a. /ʔáyo-én/ → [ʔaywén] ‘cheer-up GOAL-FOCUS’
 - b. /babáwi-én/ → [babawyén] ‘regret GOAL-FOCUS’

Icelandic (Games, 1976) and Classical Nahuatl (Andrews, 1975) (both cited in Hansen & Myers 2014) are examples of languages which, like Japanese, lack length contrasts in glides. See Hansen & Myers (2014) and Kawahara et al. (2011) for other relevant examples from other languages. The rest of this paper focuses on Japanese, although there is no reason not to extend the overall conclusion to other languages.

(Turk et al., 2006; Myers & Hansen, 2005). As a result, their constriction durations are hard to perceive. Since a difference in constriction duration serves as a primary cue for singleton-geminate contrasts (e.g. Kawahara 2015 for a recent review), singleton-geminate distinctions are hard to distinguish for sonorants.

For the sake of illustration, Figures 1 and 2 provide waveforms of singleton-geminate contrasts in stops and glides in Arabic (Kawahara, 2007). While stops have clear boundaries with the surrounding vowels, glides have very blurry boundaries. It is therefore difficult to know where the glides begin and where they end—this is a general problem that anybody who has tried to segment a glide from an adjacent vowel in an acoustic study will face. For this reason, it is expected that the constriction durations are harder to accurately perceive for sonorants than for obstruents.

Besides the acoustic blurriness of segmental boundaries of sonorants, another factor that may work against the accurate perception of duration of sonorants is that changes in amplitudes—or changes in perceived loudness—facilitate the detection of segmental boundaries (Kato & Tsuzaki, 1994; Kato et al., 1997). Because sonorantal boundaries with spectral continuity involve less amplitude/loudness changes than obstruent boundaries, sonorants have yet another disadvantage in signaling their boundaries. This issue is more fully addressed in Experiments III and IV.

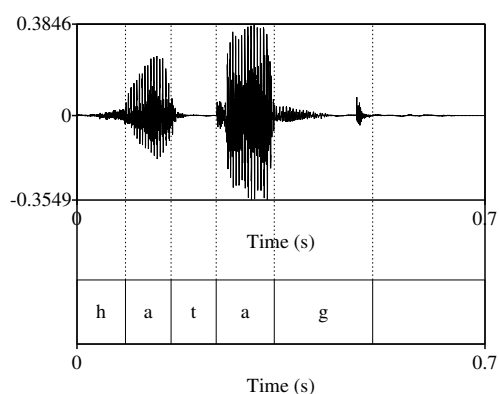


Figure 1: Arabic [t]-[tt] pair.

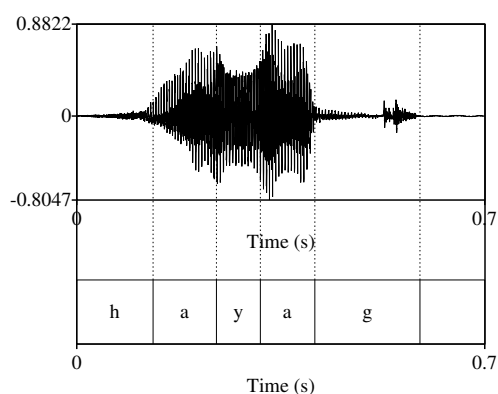


Figure 2: Arabic [y]-[yy] pair.

As summarized here, Podesva (2002) offers an interesting and plausible story about the grounding of the dispreference against sonorant geminates. However, no perception experiments have been reported to test this hypothesis. Partly to address this problem, Kawahara (2007) created continua from geminates to singletons for each type of geminates in Arabic, and presented them to Arabic speakers for an identification task. The results show that the identification functions were steeper for obstruents than for sonorants—more of the continuum was consistently categorized for obstruents than for sonorants. However, the relationship between the steepness of identification functions and the distinctiveness of singleton-geminate contrasts does not seem straightforward to interpret. Moreover, the experiment used speech sounds of Arabic as stimuli and Arabic listeners as participants. Therefore, the effect of factors other than sonority—such as lexical frequencies of each type of geminates or transitional probabilities from preceding consonant to each of singletons and geminates—remained unclear, and possibly worked as confounds.³

Experiments I and II reported below more directly test the relative non-distinctiveness of singleton-geminate contrasts in sonorants. To control for phonetic factors other than spectral continuity, the experiments used non-speech sounds that mimicked singleton-geminate contrasts in obstruents and sonorants.

2.2 Some caveats

A few remarks are in order before proceeding to the description of the experiments, first on the theoretical context. Podesva's general idea is couched within the general framework of Adaptive Dispersion Theory (Liljencrants & Lindblom, 1972; Lindblom, 1986), which is incorporated into generative phonology (Flemming, 1995, 2004; Padgett, 2002, 2009) via Optimality Theory (Prince & Smolensky, 2004). Within this framework, sonorant singleton-geminate pairs can be marked because they are not perceptually distinct. In this theory, it is the singleton-geminate contrasts in sonorants, not the sonorant geminates *per se*, that are marked; see the references cited above for formal implementations of this idea. An alternative is to encode these sorts of perceptual effects on phonology through diachrony (Barnes, 2002; Blevins, 2004a,b; Yu, 2004).⁴ This paper is not intended to solve this debate about whether perceptibility should be encoded synchronically or diachronically. Instead, the aim of Experiments I and II is to test the assumption behind Podesva's (2002) hypothesis—the non-distinctiveness of sonorant singleton-geminate pairs—but we do not commit ourselves to any particular theoretical implementation of this idea.

Second, the confusability problem between singletons and geminates may not be the only

³Various lexical factors impact the categorization of speech sounds, which include the distinction between word vs. non-word (Ganong, 1980), lexical frequency differences (Connine et al., 1993), neighborhood densities (Vitevitch & Luce, 1999), and transitional probabilities (McQueen & Pitt, 1996).

⁴Though see Kawahara (2006), Hayes & Steriade (2004), Hura et al. (1992), Martin & Peperkamp (2011), Moreton (2008), Steriade (2008), Wilson (2006), and Zsiga (2011) for arguments for encoding phonetic factors in synchronic phonological systems.

source of the avoidance of sonorant geminates. For example, given intervocalic geminate glides (e.g. [iyyi]), it is conceivable that the first part of the geminate can be confused as a part of a preceding (long) vowel (cf. Myers & Hansen 2005). Also concerning rhotic geminates, it would be impossible to prolong the duration of a tap or a flap, and they would instead have to turn into a trill, in order to become a geminate while keeping its rhoticity. However, a trill requires a very precise articulatory coordination (Ladefoged & Maddieson, 1996; Solé, 2002). In short, we do not intend to claim that low distinctiveness of singleton-geminate pairs is the only phonetic problem for sonorant geminates.

Neither is it the case that sonorant geminates are the only kinds of geminates that are avoided for a phonetic reason. For example, voiced obstruents geminates are known to be avoided in many languages because it is difficult to maintain voicing during obstruents for a long stretch of time for an aerodynamic reason (Hayes & Steriade 2004; Ohala 1983; Westbury & Keating 1986).⁵

3 Experiment I: Discrimination experiment, obstruents vs. sonorants

3.1 Introduction

This experiment tested whether sonorantal spectral continuity makes a short-long pair difficult to distinguish. The stimuli were non-speech analogues mimicking singleton-geminate pairs of stops, fricatives, and sonorants. The experiment used non-speech stimuli so as to control for acoustic parameters other than spectral continuity, such as preceding vowel duration, intensity of surrounding vowels, and duration of consonant intervals themselves. In experiments using real speech, on the other hand, it is difficult to control for the duration of consonant intervals because the duration of glides is difficult to measure, for reasons stated in section 2 (see also Turk et al. 2006). Using non-speech sounds also avoided perceptual bias effects, such as lexical bias (Ganong, 1980), lexical frequency bias (Connine et al., 1993) or transitional probability bias (McQueen & Pitt, 1996).

3.2 Method

3.2.1 Stimuli

The three conditions were non-speech analogues of stops, fricatives, and sonorants. All the stimuli had VCV structure in which the duration of C was varied. Non-speech analogues of vowels consisted of anharmonic complexes of sine waves (Kingston et al., 2009). They consist of 50 sine waves ranging from 100Hz to 16kHz and separated by equal natural log intervals. The amplitude

⁵It may as well be the case that spectral continuity at low frequency range in voiced stops makes the perception of duration harder for voiced stops than for voiceless stops. However, this paper sets this hypothesis aside, because the aerodynamic challenge of voiced stops geminates is well-established.

of each sine wave negatively correlated with its frequency. The peak amplitude of the anharmonic complexes was set to 0.8(Pa) by Praat (Boersma, 2001; Boersma & Weenink, 1999–2014).

Consonant intervals mimicked the acoustic properties of stops, fricatives and sonorants; i.e., silence for stops, white noise filtered between 2kHz and 22kHz for fricatives, and the same interval as the vocalic interval with half of its peak energy for sonorants. Figures 3 and 4 illustrate the parallels between the non-speech stimuli and the corresponding speech forms. The duration of the short consonants were set to 100ms and that of long consonants were set to 150ms. These two values were chosen because the short-long contrasts based on these values were neither too easy or too difficult to discriminate in a pilot study. The vocalic intervals were set to be 100ms. For the discrimination experiment, two VCV sequences were concatenated with 400ms inter-stimulus intervals (ISI).

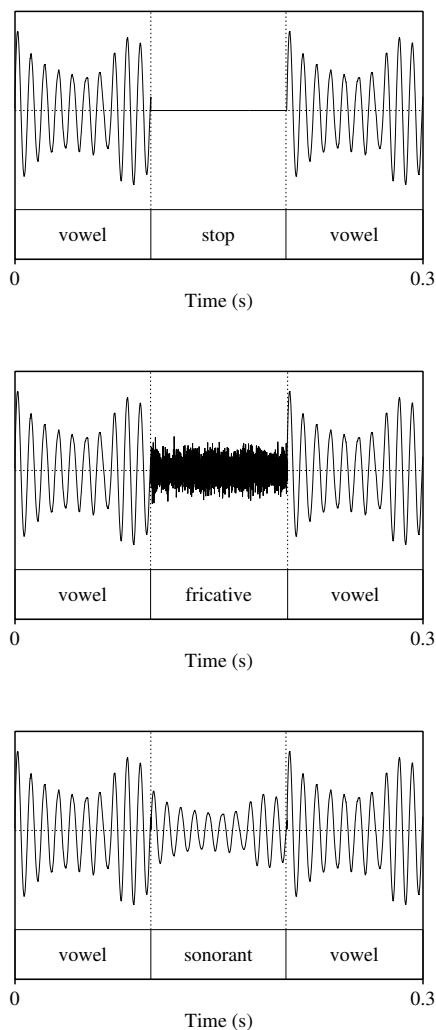


Figure 3: The stimuli. Top=stop; middle=fricative; bottom=nasal.

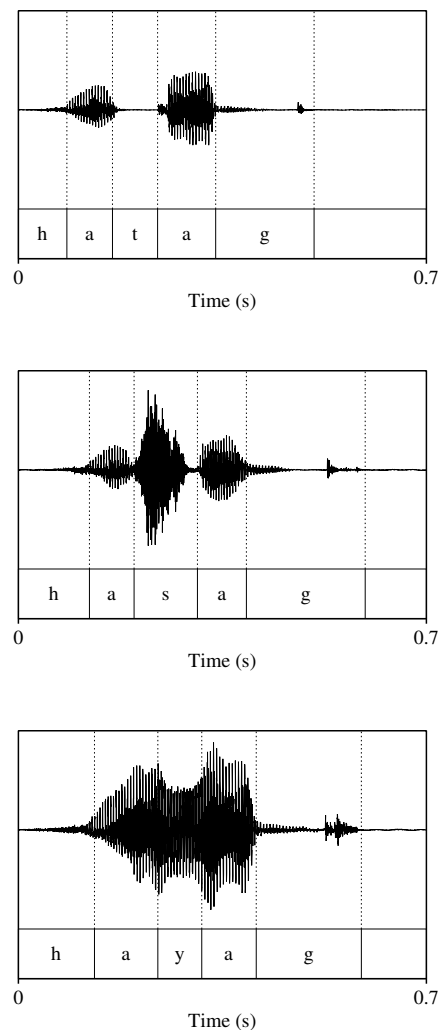


Figure 4: Corresponding speech forms (in Arabic).

3.2.2 Procedure

The task was a same-different discrimination (AX-discrimination) experiment. Four pairs of combinations of S(hort) and L(ong) stimuli—SS (same), LL (same), SL (different), LS (different)—were created for each condition. Participants went through all the stimuli once in the practice block while receiving feedback. An experimenter stayed with the participants during the practice run so that if the participants had remaining questions, they could be answered.

The main session presented 25 repetitions of all the stimuli, thus a total of 300 pairs (25 repetitions * 4 same-different pairs * 3 conditions). The participants kept receiving feedback during the main session in the form of the correct answer (i.e. Same or Different). Superlab (ver 4.0) was used to present the stimuli and feedback (Cedrus Corporation, 2010). The order of the stimuli was randomized. All the participants wore high quality headphones (Sennheiser HD 280 Pro), and registered their responses using a Cedrus RB-730 response box. The experiment took place in a sound-attenuated laboratory.

3.2.3 Participants

Twenty-five native speakers of English participated in this experiment. They received course credit for their classes for participating in the experiment.⁶ English does not have singleton-geminate contrasts, and hence their native language knowledge should not make one particular singleton-geminate contrast easier to discriminate than the other contrasts.

3.2.4 Analysis

d' -values were deployed as a measure of discriminability to tease apart sensitivity from bias.⁷ Given the roving mode of the experiment in which different types of pairs were presented in one session, a differentiating mode of discrimination was assumed (Macmillan and Creelman 2005: 221-225). d' -values were calculated using `psyphy` package (Knolauch, 2009) of R (R Development Core Team, 1993–2014). In a few cases, hit rates were lower than false alarm rates. In that case, negative d' -values were replaced with zero. Two listeners showed lower hit rates than false alarm rates in two out of three conditions, so their data was excluded. d' -values across three conditions were compared using a within-subject t-test. The alpha level was Bonferroni-adjusted according to the number of comparisons (.05/3=.017).

⁶The participants of all the four experiments have received extra course credit, and hence this information is not repeated below.

⁷Using d' -values is particularly important in this experiment, because in AX-discrimination tasks, listeners are often biased to saying “Same”, unless they hear a clear difference.

3.3 Results

Figure 5 illustrates the results of Experiment I. Each scatterplot compares d' -values in two different conditions. Each point within a scatterplot represents the pair of d' -values for each participant. Any point that is to the left of the diagonal axis shows that the listener had a higher d' -value for the condition represented in the y-axis; any point that is to the right of the diagonal axis shows that the listener showed a higher d' -value for the condition that is represented in the x-axis.

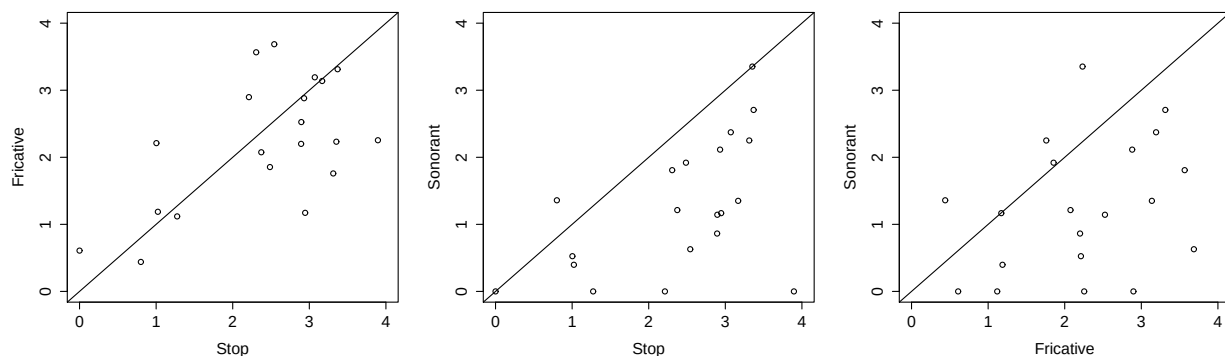


Figure 5: The distribution of d' -values in each condition in Experiment I.

In the stop-fricative comparisons, some listeners showed higher d' -values in the stop condition, while others showed the opposite pattern. The stop condition and the fricative condition thus did not differ significantly (the averages: stop 2.63 vs. fricative 2.25; $t(22) = 1.95, n.s.$). In the other two panels, most, if not all, listeners showed lower d' -values in the sonorant condition than in the stop or the fricative conditions (the average for the sonorant condition=1.38). Statistically, the sonorant condition was different from the stop condition ($t(22) = 5.71, p < .001$), and the fricative condition ($t(22) = 3.56, p < .01$).

3.4 Discussion

The result shows that sonorantal spectral continuity does make the short-long pair less discriminable. This result supports Podesva's (2002) hypothesis that sonorantal spectral continuity makes the duration of the consonantal intervals difficult to distinguish, and hence make the short-long pair harder to discriminate.

Admittedly the current experiment, or any experiment for that matter, cannot prove the *causal-ity* relationship between the low discriminability of a durational contrast of spectrally continuous intervals and the fact that Japanese avoids sonorant geminates. However, the experiment does show the *correlation* between the two observations. It therefore seems reasonable to speculate that the avoidance of sonorant geminates may have its root in the discriminability problem of the

singleton-geminate contrasts, in some way or another.

4 Experiment II: Identification experiment, obstruents vs. sonorants

4.1 Introduction

Experiment I shows that it is hard to distinguish a short-long pair when the consonant interval is spectrally continuous with surrounding vocalic intervals. Experiment II followed up on this result with an identification experiment, which addressed whether spectral continuity makes it challenging to learn the short and long category. Although a discrimination experiment has an advantage in that the participants do not need to learn two categories, an identification experiment emulates the language acquisition situation more closely. During the course of acquisition, language learners need to learn the short and long categories based on tokens presented in isolation.

4.2 Method

4.2.1 Procedure

The identification experiment used the same set of stimuli as Experiment I. Listeners learned two categories in the practice phase, and were tested on how well they learned each category in three different conditions. Listeners were not told that the two categories were based on durational differences; instead the short category was labeled as A and the long category was labeled as B—this format again emulates the actual language acquisition situation: learners are not explicitly told that a singleton-geminate contrast is based on differences in duration.

Since a pilot experiment showed that it is difficult to learn the two categories for three types of non-speech sounds at the same time, each type of stimuli (stop, fricative, and sonorant) was blocked into small, separate sessions, each with its own practice phase and testing phase. Since the order of learning these three categories might influence their performance, the order of the presentation of the three blocks was controlled by a Latin Square design. Group 1 went through the experiments in the order of stop, fricative, and sonorant; Group 2 in the order of fricative, sonorant, and stop; Group 3 in the order of sonorant, stop, fricative.

The practice session consisted of three phases. The first phase presented five repetitions of A-B chains, followed by five repetitions of B-A chains. The second phase presented five repetitions of A in isolation and five repetitions of B in isolation. In the final practice phase, the participants were tested on 15 tokens of each with feedback. A main session contained 60 tokens of each of the short and long stimuli. The order of stimuli was randomized during the main sessions. Feedback was provided in the main session as well, because a pilot experiment without feedback resulted in performances near chance. All other aspects of the experimental methodology are the same as Experiment I.

4.2.2 Participants

Eight native English speakers participated in each Latin Square order (a total of twenty-four speakers).

4.2.3 Analysis

As with the discrimination experiment, d' -values were used as a measure of sensitivity. Three listeners showed a negative d' -value in one of the three conditions; these values were replaced by 0. One listener showed negative d' -values for two out of the three conditions, and this person's data was therefore thrown out. Another listener was run to compensate for the gap. d' -values in the three conditions were compared using a within-subject t-tests. Since the predictions were clear from the results of Experiment I, the alpha-level was not adjusted.

4.3 Results

Figure 6 illustrates the distribution of d' -value for each listener in Experiment II.

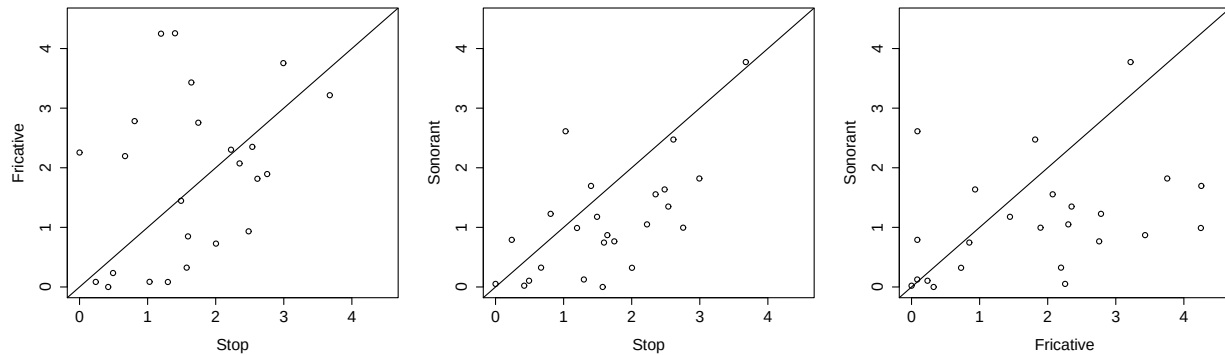


Figure 6: The distribution of d' -values in each condition in Experiment II.

As was the case for Experiment I, some listeners were better at the stop condition, while others were better at the fricative condition; therefore, there was no significant difference between these two conditions (the averages: stop=1.63 vs. fricative: 1.84; $t(23) = -0.73, n.s.$). On the other hand, d' -values for the sonorant condition were generally lower than those for the stop condition (the middle panel: $t(23) = 3.29, p < .01$) or those for the fricative condition (the right panel: $t(23) = 2.68, p < .05$) (the average for the sonorant condition=1.10). In terms of the order effect, the average d' -values increase in successive blocks (1st block: 1.35; 2nd block: 1.47; 3rd block: 1.71), although this correlation did not reach significance ($\rho = .14, n.s.$).

4.4 Discussion

The results show that the short and long categories are generally harder to learn for the sonorant condition than the obstruent conditions. There was one listener who showed a very high d' -value in the sonorant condition (2.61) compared to the stop (1.03) or the fricative condition (0.09). This listener took the sonorant condition in the third block; therefore, it may be that this listener got used to identifying non-speech stimuli after the first two blocks.⁸ All the other listeners showed a d' -value for the sonorant condition that is lower than or comparable to the d' values for the other two conditions. These results show that a duration contrast that is spectrally continuous with surrounding intervals is harder to learn than contrasts that are spectrally not continuous.

The results of Experiments I and II may thus offer a phonetic explanation for why Japanese avoids sonorant geminates.

5 Experiment III: Discrimination experiment, nasals vs. glides

The next two experiments addressed the distinction between nasal geminates and glide geminates. Experiments III and IV pursued a hypothesis that this effect of sonority derives from the fact that amplitude changes facilitate the perception of segmental boundaries (Kato & Tsuzaki, 1994; Kato et al., 1997).⁹ Given VCV sequences, larger amplitude changes in both VC- and CV-transitions should facilitate the demarcation of consonantal boundaries. If so, sonorants with high sonority (for example, glides) have a disadvantage in signaling their edges with respect to the surrounding vowels. In other words, glides have the problem of not having large amplitude changes, in addition to the blurriness problem identified in Experiments I and II. To test this hypothesis, Experiments III and IV tested whether larger amplitude drops facilitate the categorization and discrimination of singleton-geminate contrasts.

5.1 Method

5.1.1 Stimuli

As with Experiments I and II, Experiment III used non-speech sine waves to control factors other than amplitude drops. All the stimuli mimicked VCV structures, as illustrated in Figures 7 and 8. All components were made out of pure sine waves, but with varying amplitudes. The vocalic intervals were 100ms in duration and were presented at 70dB. The consonants were either short

⁸Recall that the order effect did not reach statistical significance. Therefore, this learning effect must have been strong specifically for this participant.

⁹Recall from section 2.2 that gemination of rhotics is likely to also involve, in addition to the perceptual problem hypothesized here, articulatory difficulties: it is articulatorily difficult to lengthen a tap because a tap involves a short closure in the first place, and a trill faces its own articulatory/aerodynamic difficulty (Ladefoged & Maddieson, 1996; Solé, 2002). A clear comparison can still be made between nasals and glides.

(80ms) or long (145ms) with 10ms of transition on each side. In one condition, the consonant was 64dB (the 6dB drop condition, Figure 7) and in the other condition the consonant was 52dB (the 18dB drop condition, Figure 8).

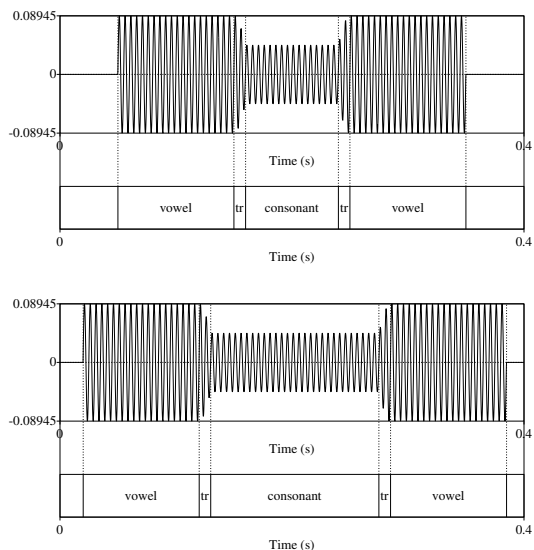


Figure 7: The 6dB drop condition. The top panel=short; the bottom panel=long.

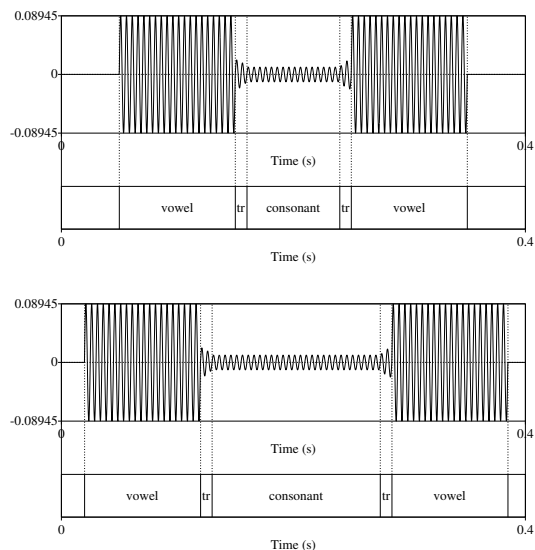


Figure 8: The 18dB drop condition. The top panel=short; the bottom panel=long.

5.1.2 Other procedure

The details of the procedure were identical to those of Experiments I and II, other than those noted here. The task was a same-different discrimination experiment. The experiment had four pairs of combinations of S(hort) and L(ong) stimuli—SS, LL, SL, LS—for each of the two condition. The ISI was 400ms. Superlab was used to present the stimuli and feedback in the form of the correct answer.

Participants first went through all the stimuli once in the practice block while receiving feedback. The order of the stimuli was randomized. The main session presented 50 repetitions of all the stimuli, thus a total of 400 pairs (50 repetitions * 4 same-different pairs * 2 amplitude change conditions). The order of the stimuli was randomized in the main session, and the participants kept receiving feedback during the main session as well.

Twenty-three native speakers of English participated in this experiment. To analyze the results, d' -values were calculated as a measure of discriminability in the same way as Experiment I.

5.2 Results and discussion

Figure 9 illustrates the results of the discrimination experiment. The scatterplot compares d' -values in the two different conditions. Each point within a scatterplot shows a pair of d' -values for each participant. Any point that is to the right of the diagonal axis shows that the listener had a higher d' -value in the 18dB drop condition. Almost all listeners showed higher degree of sensitivity to a short/long contrast in the 18dB drop condition.

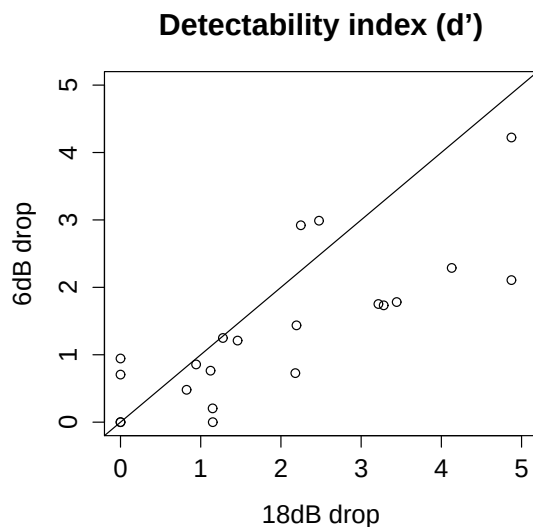


Figure 9: The distribution of d' -values in Experiment III.

The average d' -values were statistically higher for the 18dB drop condition than the 6dB drop condition (1.94 vs. 1.35, $t(22) = 5.54, p < .001$). Experiment III thus shows that larger amplitude changes do facilitate the discrimination of singleton-geminate contrasts. As we interpret the results in terms of actual speech, short/long contrasts should be less perceptible for glides than for nasals: glides involve less of amplitude changes from surrounding vowels, which would make the perception of their duration—and consequently their short-long contrasts—difficult to perceive. This result matches well with the actual phonological observation that Japanese avoids glide geminates more than nasal geminates.

6 Experiment IV: Identification experiment, nasals vs. glides

Experiment III shows that it is harder to distinguish a short-long pair when the consonant intervals involve smaller amplitude changes with respect to the surrounding vocalic intervals. Experiment IV followed up on this result with an identification experiment, which addressed whether smaller amplitude changes make it more challenging to learn the short and long categories.

6.1 Method

Experiment IV used the same set of stimuli as Experiment III. However, in this experiment, VCV tokens were presented in isolation, as with Experiment II. Listeners learned two categories of the consonant interval (short or long) in the practice phase, and were tested on how well they learned each category in the two different conditions. The short category was labeled as A and the long category was labeled as B.

Each type of stimuli (the 6dB drop condition and the 18dB drop condition) was blocked into small, separate sessions, each with its own practice phase and testing phase. The order of learning two categories was counterbalanced across the participants.

The practice session consisted of three phases. The first phase presented five repetitions of A-B chains, followed by five repetitions of B-A chains. The second phase presented five repetitions of A in isolation and five repetitions of B in isolation. In the final practice phase, the participants were tested on 15 tokens of each with feedback. The main session contained 90 tokens of each of the short and long stimuli. The order of stimuli was randomized during the main sessions. Feedback was provided in the main session, as was the case with Experiment II.

Twenty English speakers participated in this experiment.

6.2 Results and discussion

Figure 10 illustrates the distribution of d' -value for each listener in Experiment IV.

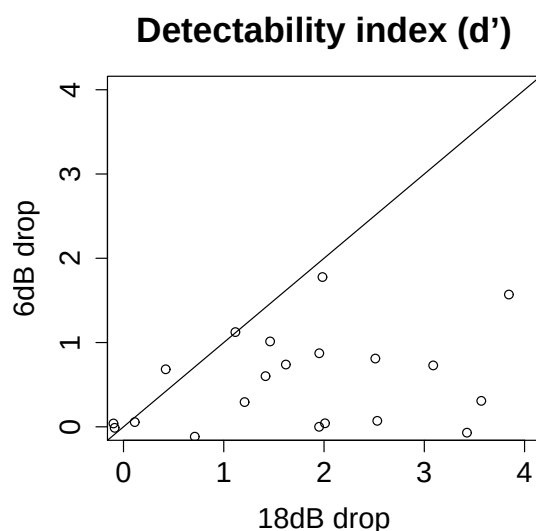


Figure 10: The distribution of d' -values in Experiment IV.

Listeners learned the contrast between short and long contrasts better in the 18dB drop condi-

tion. Almost all listeners showed higher d' -values in the 18dB drop condition, and the difference between the two conditions was significant (averages: 1.74 vs. 0.53, $t(19) = 4.68, p < .001$). The results thus show the short and long categories are easier to learn when there are more amplitude drops.

The results of Experiment III and IV show that larger amplitude changes facilitate both discrimination and categorization of a short-long contrast. The results accord well with the claim that amplitude changes facilitate perceptual demarcation of segmental boundaries (Kato & Tsuzaki, 1994; Kato et al., 1997). The results also imply that, since more sonorous consonants (e.g. glides) involve smaller amplitude changes with respect to surrounding vowels, the singleton/geminate distinction would be harder to perceive for more sonorous consonants.

7 General discussion

This paper started with two phonological observations about Japanese geminates: (i) no glide geminates are allowed at all to make lexical contrasts; (ii) nasal geminates are allowed, but nevertheless avoided by a phonological process. The four experiments reported in this paper show that spectral continuity and lack of amplitude changes make the perception of length contrasts difficult. Glide geminates have both problems, while nasal geminates have the spectral continuity problem.

It is not possible to prove the causality relationship between the difficulty of perception of particular geminate types and the phonological behavior in Japanese. However, the experimental results are at least compatible with what we would expect if languages disprefer contrasts that are hard to perceive (Flemming, 1995; Padgett, 2002; Liljencrants & Lindblom, 1972; Lindblom, 1986). It is also beyond the scope of this paper to argue how direct this relationship might be (to the extent that there is a relationship). It could be that this relationship is very direct and synchronic (Flemming, 1995; Padgett, 2002), or indirect and diachronic (Blevins, 2004a,b).

Regardless of how conclusive we can be about how the phonetics-phonology interface works, the descriptive values of this paper remain the same—when presented with non-speech analogue of short-long pairs, speakers were better at learning the length distinction if (i) they involve spectral discontinuity, and (ii) more amplitude changes with respect to surrounding vowels.

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